8

Of Snowflakes and Spacetime

SYMMETRY AND THE EVOLUTION OF THE COSMOS

Richard Feynman once said that if he had to summarize the most important finding of modern science in one sentence he would choose “The world is made of atoms.” When we recognize that so much of our understanding of the universe relies on the properties and interactions of atoms—from the reason that stars shine and the sky is blue to the explanation for why you feel this book in your hand and see these words with your eyes—we can well appreciate Feynman’s choice for encapsulating our scientific legacy. Many of today’s leading scientists agree that if they were offered a second sentence, they’d choose “Symmetry underlies the laws of the universe.” During the last few hundred years there have been many upheavals in science, but the most lasting discoveries have a common characteristic: they’ve identified features of the natural world that remain unchanged even when subjected to a wide range of manipulations. These unchanging attributes reflect what physicists call symmetries, and they have played an increasingly vital role in many major advances. This has provided ample evidence that symmetry—in all its mysterious and subtle guises—shines a powerful light into the darkness where truth awaits discovery.

In fact, we will see that the history of the universe is, to a large extent, the history of symmetry. The most pivotal moments in the evolution of the universe are those in which balance and order suddenly change, yielding cosmic arenas qualitatively different from those of preceding eras. Current theory holds that the universe went through a number of these transitions during its earliest moments and that *everything* we’ve ever encountered is a tangible remnant of an earlier, more symmetric cosmic epoch. But there is an even grander sense, a metasense, in which symmetry lies at the core of an evolving cosmos. Time itself is intimately entwined with symmetry. As will become clear, the practical connotation